BEGINNER GUIDE

Rendering a car in Blender for a website

What rendering is and why we need it, how to do it by hand in Blender, the exact commands I ran, the exploded “take it apart” animation, and how the website turns the pictures into scrolling 3D.

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1 Nov 2026 · Car: N°77 GR Supra GT300 by vecarz (CC BY 4.0)

01 What “rendering” means

A 3D model is only **shape + materials**: the surface of the car and what each part is made of (paint, glass, rubber). It isn't a picture yet. **Rendering** is the step where the computer works out how light from the lamps and the room bounces off every surface and into a camera — and turns that into a flat image.

	Real-time rendering (in the browser)	Offline rendering (Blender Cycles)
Time per picture	about 1/60 of a second	about 30–90 seconds
How light is computed	shortcuts and approximations	simulates real light rays bouncing (path tracing)
Reflections, glass, soft shadows	rough or faked	physically correct — looks like a photo
Can the visitor rotate freely?	yes	only through the angles we rendered
Works on phones?	struggles with big HD models	yes — it's just images

A browser must draw 60 pictures per second, so it can't spend a minute on one. That's why the first AETHER car (drawn live) looked dated. Instead we **render 120 film-quality pictures ahead of time**, one every 3° around the car, and the website shows the right one for how far you've scrolled. This is how Apple and many car brands get their photographic product pages.

> Remember: the model decides how much detail exists; the rendering decides how real the light looks. You need both.

02 The whole pipeline in one picture

```
  1. Find a model         Sketchfab → check licence + detail → download .glb
          |
  2. Build a studio        Blender: car + studio light (HDRI) + 2 light strips + dark floor + camera
          |
  3. Render                Cycles on the Mac GPU → 120 PNGs (turntable) + 4 close-ups + 48 exploded
          |
  4. Shrink for the web    PNG → WebP (1920 px for laptops, 960 px for phones), ~8 MB per set
          |
  5. Play on scroll        canvas draws frame = round(scroll progress × 119)
```





Three of the 120 turntable frames: 0°, 90° and 180°.

03 Choosing a model (licences matter)

Model we looked at	Verdict	Why
McLaren P1 “from Assetto Corsa”	❌ practice only	ripped from a video game — the uploader can't license it
Bugatti EB110 (Forza parts)	❌ practice only	also game parts, NonCommercial, real brand
Khronos “Car Concept”	✅ clean (CC BY)	made by the artist, but simpler body — looks like a concept render
Toyota GR Supra GT300 by vecarz	✅ chosen	CC Attribution, made by vecarz (kadir kandemir), 357k triangles, 4K textures

- Licence must say **CC Attribution** or **CC0** (not NonCommercial) for a public site.
- The uploader must be the maker. Avoid “Forza / Assetto Corsa / GTA / Gran Turismo / ripped”.
- Look for **150k+ triangles**, separate glass/chrome/rubber materials, 2K–4K textures.
- Download the **original .glb** — converted formats can lose detail.
- Credit exactly what Sketchfab's “Copy credits” gives you (it's in the site footer).





Same studio, three models: McLaren (great but ripped), Car Concept (clean but simple), the Supra (clean and HD).

04 Do it yourself in Blender — the turntable

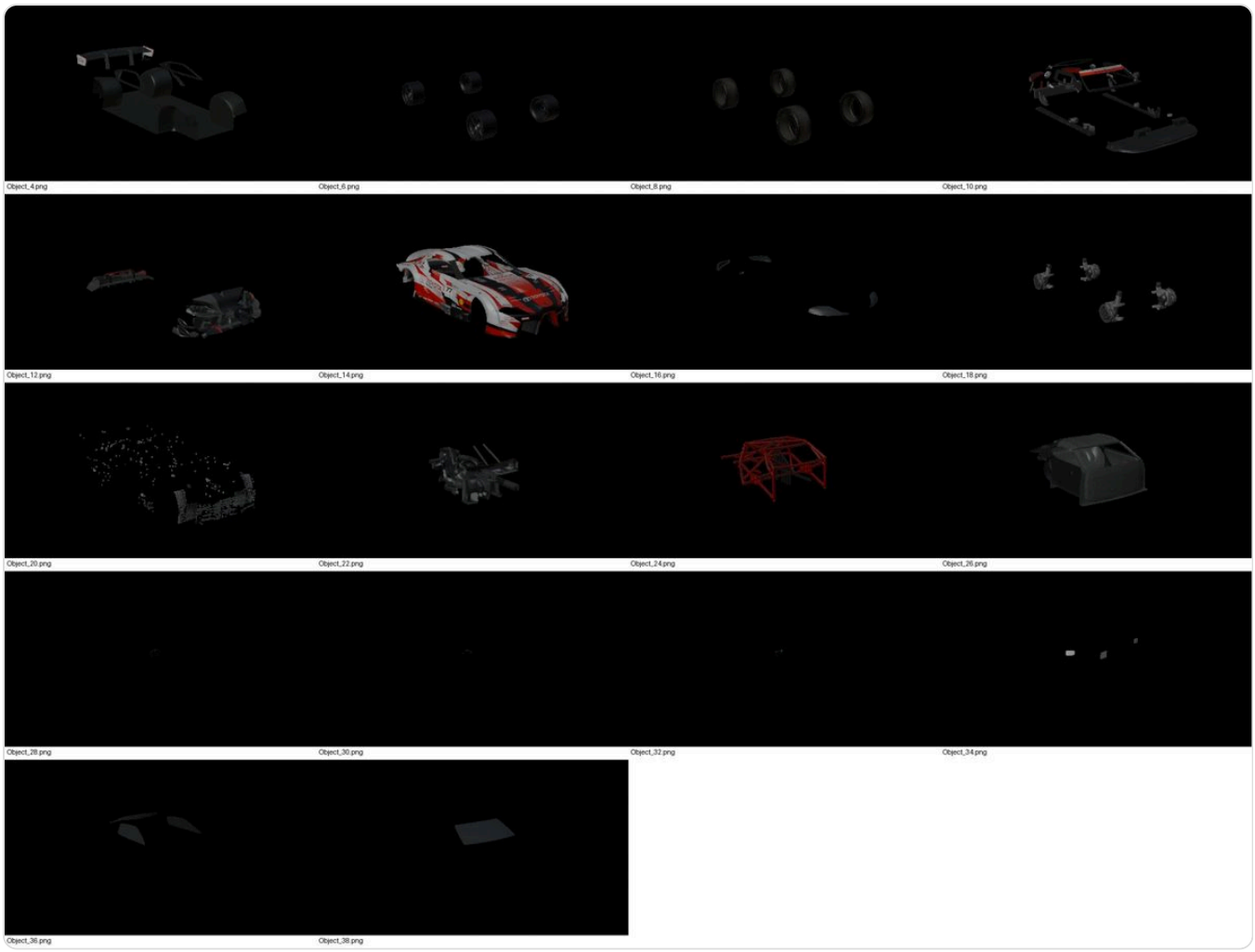
Blender is free (blender.org). These steps build exactly what my script builds. Each step teaches one idea.

Step	Where in Blender	What to do
1. Fresh scene	File → New → General	Delete the default cube, light and camera (select, press X).

Step	Where in Blender	What to do
2. Import the car	File → Import → glTF 2.0	Pick toyota_gr_supra_gt_gt300.glb. It arrives at real size.
3. Studio light	World tab (red globe) → Color → Environment Texture	Open studio_4k.hdr (Poly Haven). This lights the car and gives reflections. Strength ≈ 0.9.
4. Dark background	World node editor (Shading workspace)	Mix Shader: Light Path “Is Camera Ray” → camera sees near-black, the car still gets the HDRI light.
5. Floor	Add → Mesh → Plane, press S, type 40	Material: Principled BSDF, Base Color very dark, Roughness ≈ 0.38 (a soft reflection).
6. Light strips	Add → Light → Area	Shape Rectangle, size 0.5 × 7.5 m, Power 900 W, 4.2 m above the car. Make two, left and right. These draw the shiny lines on the paint.
7. Light only the car	Select a strip → Object tab → Shading → Light Linking	Receiver collection = a collection with just the car. The floor stays dark and moody.
8. Rim light	Add → Light → Area (violet)	Behind the car, aimed at it — a coloured edge separates the car from the background.
9. Camera	Add → Camera	Focal length 52 mm. Object Constraints → Track To → target an Empty at the car's middle.
10. Turntable	Add → Empty; parent the camera to it (Ctrl P)	Frame 1: rotation Z 0°, press I. Frame 120: Z 360°, press I. In the Graph Editor set interpolation to Linear (constant speed).
11. Render engine	Render tab	Engine: Cycles · Device: GPU (Preferences → System → Metal) · Samples 64 · Denoise ON.
12. Colour	Render tab → Color Management	View Transform: AgX · Look: Medium High Contrast (filmic, no blown-out whites).
13. Output	Output tab	1920 × 1080 · folder for frames · File Format PNG.
14. Test, then render all	F12, then Render → Render Animation	F12 renders one frame to check. Ctrl F12 renders all 120 (about 3 hours on your Mac).

> Tip: press Numpad 0 to look through the camera, and Z → Rendered for a live (slow) preview while you adjust lights.

05 The exploded “take it apart” animation



Every piece of the model rendered on its own to find out what it is: body shell, roll cage, cockpit, engine, wheels, splitter…

Part (object)	Moves	Starts at
Body shell + windows (Object_14, 36, 38, 16, 20)	up 2.7 m	0%
Roll cage (Object_24)	up 1.35 m	12%
Cockpit tub + dash (Object_26, 28–34)	up 0.45 m	22%
Engine + front bay (Object_22, 12)	forward 0.35, up 0.2	30%
Splitter / aero (Object_10)	forward 0.9 m	18%
Tyres / rims / brakes (Object_8, 6, 18)	sideways 1.25 / 0.85 / 0.45 m	8 / 16 / 24%
Floor + wing (Object_4)	stays	—

By hand: the four tyres are one object, so first split them — select it, Tab into Edit Mode, press **P** → **By Loose Parts**. Then for each group: at frame 1 press **I** → **Location**, move it (G, then Z / X and a distance), at frame 48 press **I** again. Stagger the start frames so pieces come apart one after another. Render frames 1–48 from a fixed, slightly high camera.



The fully open frame: shell on top, roll cage in red, cockpit and engine, wheels pulled out, splitter forward.

06 The commands I ran (copy and use)

I run Blender without opening its window (`-b` = background) and give it a Python script (`-P`). Everything after `--` goes to the script.

```
B=/Applications/Blender.app/Contents/MacOS/Blender
CAR=~/Downloads/assets/car/supra/toyota_gr_supra_gt_gt300.glb
HDR=~/Downloads/assets/car/studio/studio_4k.hdr
S=~/Downloads/assets/car/blender/studio.py

# 1) Turntable: frames 0–119 of 120, 64 samples, 1920×1080
STUDIO_KEEP=1 STUDIO_LENS=52 $B -b --factory-startup -P $S -- $CAR $HDR
~/Downloads/assets/car/supra/frames 0 119 120 64 1920 1080

# 2) Close-up stills (front, wheel, wing, side) at 2400×1350, 128 samples
STUDIO_KEEP=1 STUDIO_STILLS=1 $B -b --factory-startup -P $S -- $CAR $HDR
~/Downloads/assets/car/supra/stills 0 0 1 128 2400 1350

# 3) The story film: hero → tear apart → tour of 6 parts → rebuild (200 frames, ~13–30 s each)
STUDIO_KEEP=1 STUDIO_STORY=1 $B -b --factory-startup -P $S -- $CAR $HDR
~/Downloads/assets/car/supra/story 0 199 200 40 1600 900

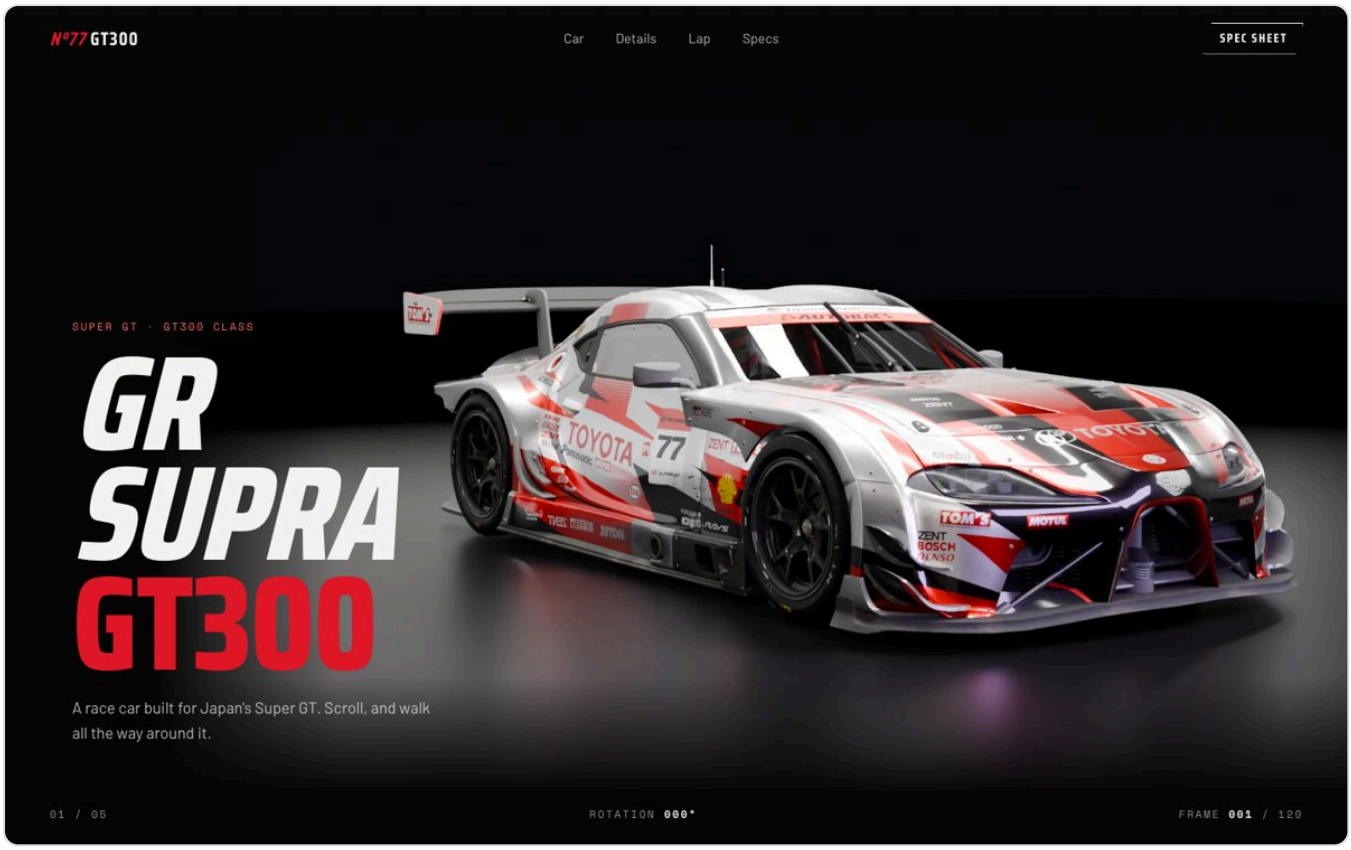
# 3b) (older) simple exploded view: 48 frames
STUDIO_KEEP=1 STUDIO_EXPLODE=1 $B -b --factory-startup -P $S -- $CAR $HDR
~/Downloads/assets/car/supra/exploded 0 47 48 64 1600 900

# 4) Shrink PNGs to WebP for the site (desktop + phone sizes)
python3 ~/Downloads/assets/car/blender/to_web.py ~/Downloads/assets/car/supra/frames
~/Downloads/aether/public
python3 ~/Downloads/assets/car/blender/to_web.py ~/Downloads/assets/car/supra/story
~/Downloads/aether/public s_ story s_
```

Setting	Meaning
STUDIO_KEEP=1	keep the model's own materials (the livery); only turn windows into real glass
STUDIO_LENS=52	camera focal length in mm — bigger = closer, flatter
STUDIO_STILLS=1	render the 4 close-up shots instead of a turntable
STUDIO_EXPLODE=1	render the simple take-apart animation instead of a turntable
STUDIO_STORY=1	render the full story: orbit, parts fly apart (shell tilts, wheels slide out), camera visits each part, everything flies home
0 119 120	first frame, last frame, total frames on the circle (120 → one every 3°)
64	samples: light rays per pixel. More = less grain, slower. 64 + denoise is plenty
1920 1080	image width and height in pixels

`queue.sh` waits for the turntable to finish, then runs the stills and the exploded view automatically. You can read every line of `studio.py` — each block has a comment saying what it builds.

07 How the website plays the pictures

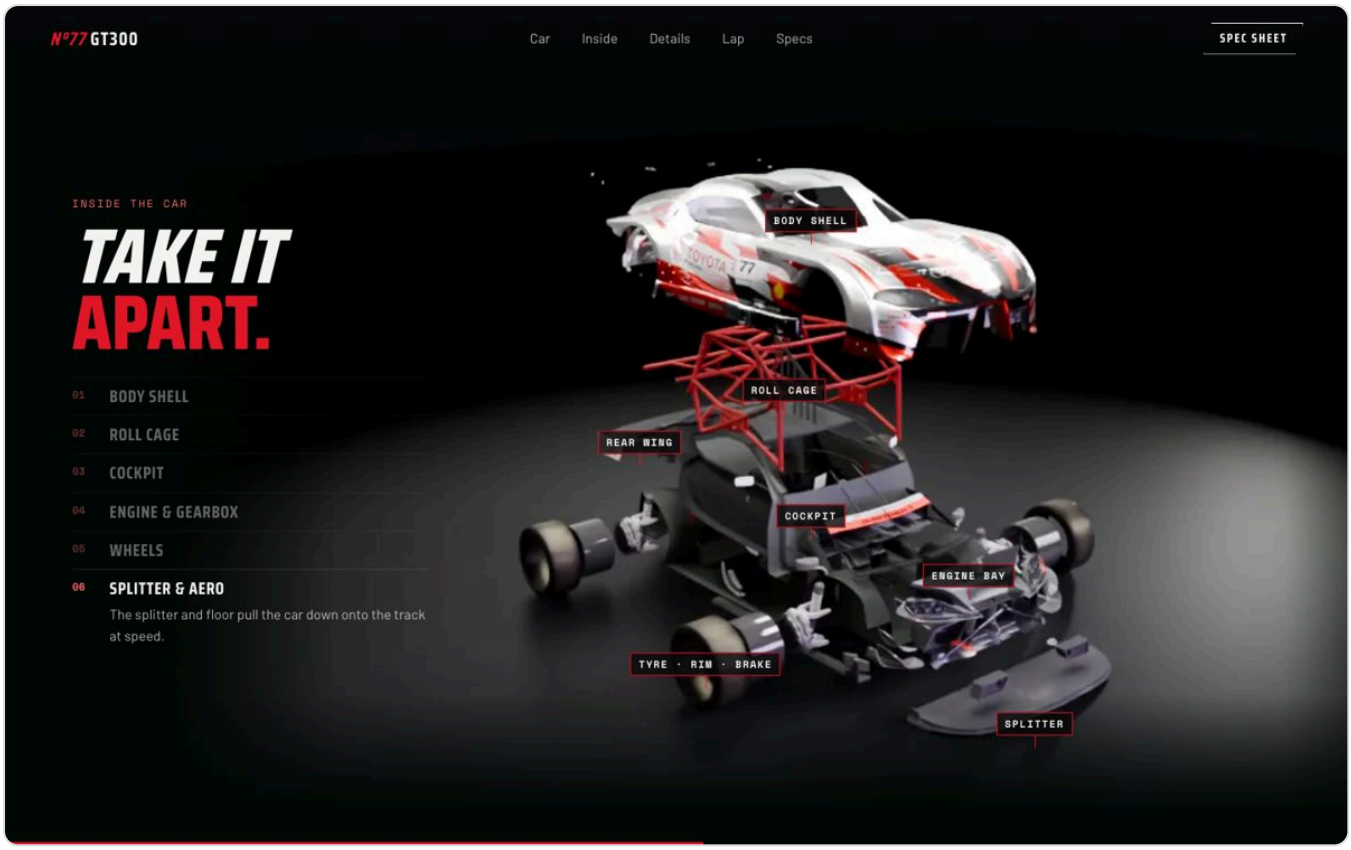


The spin: the canvas shows frame 1 of 120; scrolling turns the car.

```
// src/sequence.js (simplified)
const im = nearest(Math.round(position));      // the frame for this scroll position
g.drawImage(im, x, y, w, h);                   // draw it on the canvas
// then fade the picture's edges into the render's own background colour

// src/main.js
const p = pinProgress(spin);                   // 0 → 1 through the 560svh section
spinPos += (p * 119 - spinPos) * 0.18;         // glide toward the target frame
spinSeq.draw(spinPos);
```

- **Loading order:** every 8th frame first, then every 4th, 2nd, 1st — so the full circle works almost immediately and gets smoother as the rest arrive. The loader shows the real percentage.
- **Story section (the hero):** 200 frames in one film. Frames 0–24 orbit the whole car, 24–72 it tears apart, 72–176 the camera visits six parts (shell, cage, cockpit, engine, wheel, splitter), 176–199 it flies back together. Each text card has `data-from` / `data-to` frame numbers, so text is tied to the picture, not to scroll pixels.
- **How the story was animated in the script:** each part group has an offset, a small tilt and a start delay; the camera follows a list of key positions (overview → each part's exploded centre → overview) blended with smoothstep.
- **Phones** load half-size frames (960 px) to save data.



The exploded section: the car opens, labels point to each part, the list explains them.

08 Render times and tips

What	Time on your Mac	Size on the site
1 turntable frame (1920×1080, 64 samples)	≈ 90 s	≈ 65 KB WebP
120 turntable frames	≈ 3 hours	≈ 8 MB (desktop), ≈ 2.5 MB (phone)
4 close-up stills (2400×1350, 128 samples)	≈ 10 min	≈ 200 KB each
48 exploded frames (1600×900)	≈ 50 min	≈ 3 MB

- Test with **one frame at low resolution and 16–24 samples** before a long render.
- Grainy? Raise samples or keep **Denoise** on. Too slow? Lower resolution first (1600×900 is fine for the web).
- Keep the Mac plugged in and awake; Blender uses the GPU fully, so it gets warm — that's normal.
- If the car floats, the model has a hidden “ground shadow” sheet or stray parts — the script stands the car on its tyres.

09 Words you'll meet

Word	Meaning
Cycles / Eevee	Blender's two renderers: Cycles is realistic and slow (we use it); Eevee is fast like a game engine.
HDRI	A 360° photo of a real place, used to light the scene and give reflections.
Samples	How many light paths per pixel. More = cleaner, slower.
Denoise	An AI filter that removes leftover grain after rendering.
PBR material	Materials described physically (colour, metalness, roughness) so they look right in any light.
glTF / .glb	The web-friendly 3D file format (.glb = everything in one file).
Light linking	Telling a lamp which objects it may light (ours light only the car).
AgX	A film-like colour setting that keeps bright highlights from turning pure white.
Turntable	A camera circling an object (or the object spinning) to show every side.
Image sequence	Numbered pictures (f_000 … f_119) played like a flip-book.